Public and Private IP Addresses

Beginner-friendly networking notes for DevOps

1. Public IP Address

A **Public IP Address** is an IP address used to identify a device or network on the internet. It is generally assigned to a customer's connection by an Internet Service Provider (ISP) or network provider.

Example: 8.8.8.8 (Google Public DNS)

- Used for communication over the internet.
- Must be globally unique among currently routed public addresses.
- Can be reachable from the internet if routing and firewall rules allow it.

2. Private IP Address

A **Private IP Address** is an IP address used inside a local network, such as a home, office, or company network.

Example: 192.168.1.10

- Used for communication within a private network.
- Not directly routable over the public internet.
- Commonly assigned to computers, printers, and servers by a router or network administrator.

Private IPv4 address ranges

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

3. Difference between Public IP and Private IP

Public IP	Private IP
Used for internet communication.	Used within a private/local network.
Globally unique among routed public addresses.	Can be reused in different private networks.
Usually provided by an ISP or network provider.	Usually assigned by a router or network administrator.
May be reachable from the internet if permitted.	Not directly reachable from the public internet.
Example: 8.8.8.8	Example: 192.168.1.10

Real-world example

Internet (public IP) ↔ **Home Wi-Fi router** ↔ **Your computer** (private IP)

Your computer uses a private IP inside your home network. The router commonly uses **NAT (Network Address Translation)** to allow multiple devices to share one public IPv4 address.

Interview tip: A public IP is used for internet communication, while a private IP is used within a private network.